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Paleomagnetism and K-Ar Age of Mafic Dikes from the Wind River Range, Wyoming

ABSTRACT

Potassium-argon ages have been determined on fresh pyroxenes from three of five dikes in the Louis Lake quadrangle of the Wind River Range, Wyoming. They indicate a minimum apparent age of intrusion at 1880 m.y. This is older than published whole-rock ages from the same dikes, but is consistent with whole-rock ages of up to 2060 m.y. from other dikes in this area. The within-dike dispersion of the directions of natural remanent magnetization decreases during alternating field demagnetization up to 1400 oe. Thermal demagnetization, however, is not effective in removing secondary magnetizations. After structural correction according to overlying Cambrian sediments, an over-all mean paleomagnetic pole from four dikes is obtained at 121° W., 43° N., which is in agreement with poles from Canadian Shield rocks of similar age. There is evidence in two of the dikes for a remagnetization episode, and the equivalent pole positions suggest that this may have occurred at the time of deposition of the Cambrian Flathead Sandstone. Alteration has greatly influenced the magnetic and isotopic properties of the dikes.

INTRODUCTION

In Wyoming, large areas of Precambrian crystalline rock are exposed which form the Beartooth, Teton, Wind River, and Bighorn Ranges and which reach elevations higher than 12,000 ft. This study is concerned with some mafic dikes which intrude the core of the southern part of the Wind River Range. By isotopically dating these rocks, it was intended to compare a paleomagnetic result from mafic dikes in the American Shield with determinations from dated mafic dike swarms in the Canadian Shield (for example, Fahrig and others, 1965).

GEOLOGIC SETTING

The Wind River Range is a northwest-trending belt of high mountains, more than 100 mi long, which exposes crystalline Precambrian rocks. In the northwest sector of the Range, these are principally gray, *lit-par-lit* gneisses, extensively cut by quartz, pegmatite, and granitic rocks (Richmond, 1945; Baker, 1946). At the southern end, schists predominate and are intruded by granodiorite-quartz monzonite (Branson and Branson, 1941; Oftedahl, 1953; Parker, 1962; Hodge and Worl, 1965). These studies indicate that the final Precambrian event was the intrusion of mafic dikes.

The structure of the core is basically that of an asymmetric anticline, folded during the Laramide orogeny (Branson and Branson, 1941). Berg and Wasson (1960) and Berg and Romberg (1966) cite geophysical evidence that the southwestern limb of the anticline is overturned by a thrust fault dipping 18° to the northeast. Elsewhere on the southern flank, relations in the Precambrian core are obscured by Tertiary sediments of the Green River Basin.

The northeast flanks dip at a low angle into the Wind River Basin. From the attitude of the overlying Paleozoic sediments, this dip is variously quoted as 10° to 20° to the northeast (Branson and Branson, 1941; Berg and Wasson, 1960; Bassett and Giletti, 1963; Keefer and Van Lieu, 1970) and mapped as 15° to the northeast (Bayley, 1965). In terms of a paleomagnetic structural correction for the dikes, it is important to realize that the present dip observed in the Paleozoic strata is the result of folding. Baker (1946) emphasizes that erosion of the Precambrian continued until almost perfect peneplanation occurred, because its contact with the Middle Cambrian Flathead Sandstone is everywhere remarkably even. Thus a structural correction according to the dip of the overlying Cambrian unit is appropriate for all the directional magnetic data from the Wind River dikes.

The original attitude of the rock unit is one of the main uncertainties in a paleomagnetic study of Precambrian rocks. Nevertheless, it is

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justifiable to apply structural corrections where there is good geologic control. As an example, Prucha and others (1965) were able to demonstrate tilting of Precambrian dikes in the Bennett Mountains of the Sweetwater Uplift in Wyoming, by using structural control provided by the overlying Cambrian unit.

SAMPLING DETAILS

Because of this need for structural control, a sampling area was chosen as close as possible to the nonconformity with the Flathead Sandstone. In the Wind River Range southeast of Lander, Wyoming, the dikes have northeast trends; they are continuously exposed for several miles and they cross-cut all other Precambrian rocks (Nold, 1964; Bayley, 1965). Dike widths vary from 10 to 90 feet and in some places, the dikes appear to bifurcate, whereas in others they are offset as much as 30 ft by northwest transcurrent faults (Nold, 1964). The dikes are well exposed, for they are resistant to weathering and form long ridges (for example, Anderson Ridge).

A northwest sampling profile was made along the Louis Lake road (Fig. 1), crossing five dikes intruded into quartz monzonite. Block samples, each weighing about 15 lbs, were oriented by means of a magnetic compass at several exposures in each dike. The respective numbers of samples collected at dikes 1 to 5 were 3, 5, 5, 6, and 7. At least one sun compass bearing was taken at each dike: the difference in the two orientation methods after correcting for a regional declination of 15.5° E., (obtained from the 1965.0 Isogonic Chart, USCGS) was about 2°. The samples were distributed at intervals up to 1 mi along the length of the dike, and at different points across the dike. No fine-grained selvages were collected because they were too brittle. In the lab, two 1-in.-diameter cylinders were drilled from each block sample, and two specimens of length 1 in. were cut from the freshest part of each cylinder.

Since none of the dikes is greater than 90 ft wide, Jaeger's discussion (1957) can be used to argue justifiably that the time of cooling of any dike is very short compared with the cycle of secular variation of the magnetic field. Thus all the specimens within a dike should have been magnetized at approximately the same time, and may be given equal weight in the magnetic analysis.

ISOTOPIC WORK

An excellent summary of available isotopic ages and their implications for structural correlations in Wyoming has been provided by Giletti (1968). Bassett and Giletti (1963) described Rb/Sr and K/Ar mineral data which support an age of metamorphism of the northern end of the Wind River Range at 2700 m.y., but indicated a spread in apparent ages. This age is slightly greater than an Rb/Sr age given for muscovite from quartz monzonite in the southern part of the Range (Giletti and Gast, 1961). Bassett and Giletti (1963) infer that any age lowering may be due to hydrothermal alteration related to Laramide disturbances.

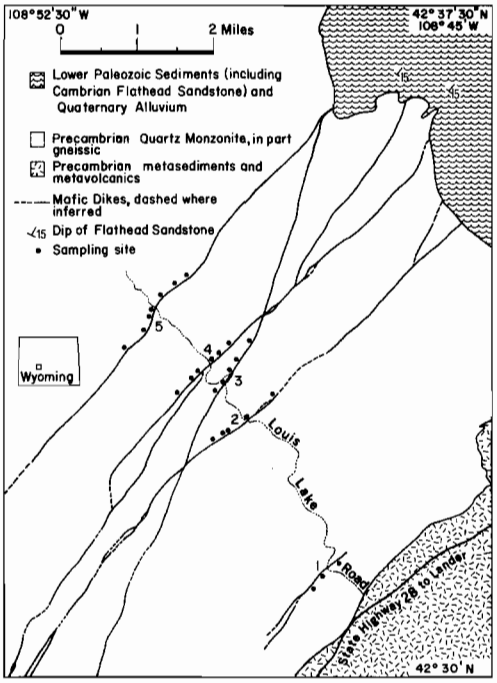


Figure 1. Partial geologic map of the Louis Lake quadrangle, Fremont County, Wyoming (*after* Nold, 1964; Bayley, 1965).

Mafic dikes are abundant in the Precambrian terranes of Wyoming, but apart from a K/Ar age of 2040 m.y. for a dike in the Bighorn Range (Heimlich and Banks, 1968), few isotopic studies have been made on the dikes. However, Paul Mueller (1970, personal commun.) is making a systematic isotopic study of dikes in the Beartooth Mountains. If several periods of dike injection were demonstrated, it would provide a reason for the resetting of other Precambrian isotopic ages in Wyoming. Condie and others (1969) describe a K/Ar dating reconnaissance of Wyoming mafic dikes, which suggests four periods of intrusion at

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TABLE 1. ANALYTICAL DATA FOR K/Ar PYROXENE AGES

Dike†	%K	Split	Sample Wt gms	Ar^{40*} $x10^{-9}$ moles	$\frac{Ar^{40*}}{Ar^{40}total}$	Age $x10^6$ yrs
2	0.1376	1.	1.802	1.281	93	1760±40
		2.	0.551	0.4277	79	1860±40
3	0.1590	1.	0.507	0.3916	70	1690±40
		2.	0.483	0.3695	62	1680±40
4	0.1364	1.	0.513	0.4011	80	1880±40
		2.	1.806	1.409	93	1880±40

† See Figure 1.

Ar^{40*} is radiogenic argon

$\lambda_\beta = 4.72 \times 10^{-10} yr^{-1}$

$K^{40}/K = 1.22 \times 10^{-4} g/g$

$\lambda_e = 5.85 \times 10^{-11} yr^{-1}$

2500, 1900 to 2200, 1400 to 1800, and 700 to 1000 m.y. Ages from the Wind River dikes fall in the last three ranges. They are all based on whole-rock data, and the samples were exclusively from fine-grained contact zones. As Leech (1966) points out, the whole-rock method must be used with care in dating older Precambrian basic intrusions, particularly those which have undergone alteration.

Petrography of the central portions of the dikes collected in the present study suggests that they are an original feldspar-pyroxene rock. Different parts have been altered to different extents. The feldspar is andesine in composition, and is saussuritized, with epidote rims. The feldspar grains are often disseminated with amphibole needles and blebs of Fe-Ti oxide, suggesting that this alteration is deuteric. Most of the pyroxenes are fresh, but some are altered to a swirling aggregate of tremolite-actinolite-biotite-Fe-Ti oxide, the alteration proceeding along cracks and inward from grain margins. Thus it was decided that more meaningful K/Ar age information might be obtained from pyroxenes, especially in parts of the dike least affected by alteration.

K/Ar ages were determined at the Mobil Field Research Laboratory, Dallas, according to techniques outlined by Burke and others (1969). Five replicate analyses were made on each of two splits of the pyroxene fraction from each sample: the analytical data are given in Table 1.

The agreement of duplicate runs on two of the samples is excellent. However, on the sample from dike 2, the 100-m.y. difference in the two determinations exceeds analytical error. This discrepancy is probably due to contamination in the potassium-poor pyroxene (only 0.1376 percent K). Nevertheless, it is believed that these pyroxene ages are the most reliable that can be determined with the available samples.

The pyroxene ages are compared in Table 2 with whole-rock ages obtained by Condie and others (1969). The interesting fact is that although the pyroxene ages are from 14 to 44 percent higher than whole-rock ages from the *same* dike, they are closely consistent with whole-rock ages from *other* similar trending dikes within 20 mi of the Louis Lake dikes. McDougall (1961, 1963) obtained very reliable K/Ar ages on pyroxenes from mafic dikes, although anomalously high ages using *metamorphic* pyroxenes are discussed by (among others) Hart and Dodd (1962), Schwartzman and Giletti (1968), and Fenton and Faure (1969). In the present study, the agreement of fresh-pyroxene ages with whole-rock ages from other Wind River dikes argues that these *igneous* pyroxene ages are meaningful. Moreover, the

TABLE 2. COMPARISON OF WHOLE-ROCK AND PYROXENE K/Ar AGES ($x10^6$ YRS)

Dike*	Width of dike (ft)	Pyroxene ages (m.y.)	Whole-rock ages (m.y.) (Condie and others, 1969)
1	10		900
2	20	1760, 1860	1410
3	90	1680, 1690	1480
4	80	1880, 1880	1300
WRD 2, 7	Not sampled		1780, 1790
WRD 43	in the		2060
WRD 47	present study		1890

* See Figure 1.

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similarity in petrology of the dikes in this part of the southern Wind River Range, coupled with a constant trend (Nold, 1964), supports the idea of synchronous intrusion. Petrographic evidence and the variation between sample splits suggest that alteration is the most critical factor in causing the large range in observed ages from these dikes.

The highest pyroxene age is 1880 m.y.; the highest whole-rock age is 2060 m.y. Because of alteration this would give a *minimum* estimate of the apparent age of 1880 to 2060 m.y. for these southern Wind River dikes.

EXPERIMENTAL METHODS

The direction and moment of the natural remanent magnetization (NRM) of all the specimens were measured on a PAR spinner magnetometer, Model SM-1. One specimen from each sample was subjected to progressive alternating field (AF) demagnetization in fields of 175, 350, 700 and 1400 oe. A second specimen from each sample was thermally demagnetized by the stepwise method in a field free space ($\pm$ 10 gamma) to 200, 300, 400, 500 and 570°C stages and the NRM was measured after the specimen had been cooled to laboratory temperatures. All the demagnetization equipment is located in a shielded room (with a residual field of 200 gamma) and is described by Helsley (1969).

In the early part of the work, an AF stage at 88 oe was also used, but with the exception of dike 5, there was little change in the NRM direction, and so this stage was omitted from subsequent studies. In addition, a thermal stage at 620°C was attempted, but it was found that with few exceptions the magnetization of the specimens became unstable (Spall, 1970) due probably to magnetic noise (Stacey, 1963, p. 99-101). Further ambitions at this temperature were therefore abandoned. As will be pointed out later, many of the directions at lower temperatures were also affected by magnetic noise.

All the directional data were corrected for the dip of 15° toward the northeast observed in the overlying Cambrian sediments, and within-dike means at each demagnetization stage were computed, using Fisher's method (1953). By this means k (the precision parameter) is obtained, which is an estimate of the dispersion of the NRM vectors; thus when k is large, the directions are tightly grouped and when k is small, they are scattered. The change in dispersion during demagnetization can be conveniently charted by the ratio k/k_0, where k_0 is the dispersion of a statistically random population at the 95 percent level of confidence (*see* Irving and others, 1961). Thus when $k/k_0 \leq 1$, the distribution may be considered to be statistically random. This method is useful for normalizing the dispersion from different populations with unequal numbers of samples. In addition, the tabulations by McElhinny (1964) allow the dispersions after successive demagnetization steps to be compared in order to test whether there has been a significant convergence of the NRM directions.

The problem of finding the demagnetizing stage for any dike which effectively removed secondary magnetizations was solved by a combination of systematic vector rotation (that is, continuous movement of the mean NRM direction along a great circle) and minimum dispersion of the individual NRM directions. Thus only the primary magnetization is assumed to remain when no further systematic rotation of the mean NRM vector occurs during demagnetization. This should also be the point at which there is maximum convergence of the specimen NRM directions, that is, their dispersion is statistically minimum at the 95 percent level of confidence.

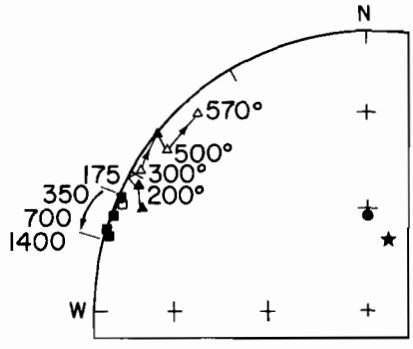
Routine measurements of the bulk susceptibility were made on a commercial bridge instrument. Strangway (1965) showed that lightning effects could be detected by high Q′ ratios, that is, the ratio of NRM intensity to susceptibility.

PILOT DEMAGNETIZATION EXPERIMENTS

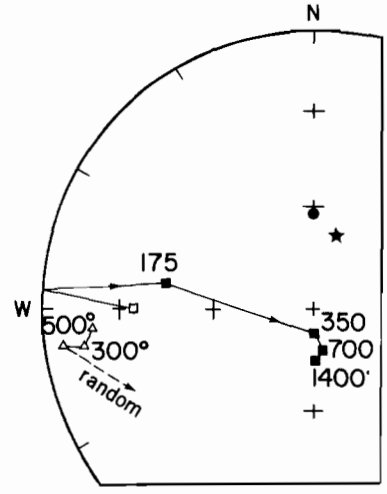
The results of the pilot AF and thermal demagnetizations are shown in Figure 2 as the mean demagnetization path for each dike, and in Figure 3 as the change in dispersion (measured by k/k_0) after each temperature or field stage. Two general comments can be made: firstly, thermal demagnetization is relatively ineffective in removing secondary magnetizations; secondly, two groups of NRM directions are present, one with steep inclinations, the other with shallow inclinations. Several detailed observations can be made:

1. For dike 1, the NRM directions are of mixed polarity very closely 180° apart: the mean has therefore been computed about a common polarity. During demagnetization, there is no change in the polarity and little change in the mean NRM directions (Fig. 2a). The dispersion significantly *decreases* during AF demagnetization (Fig. 3a) and significantly *in-*

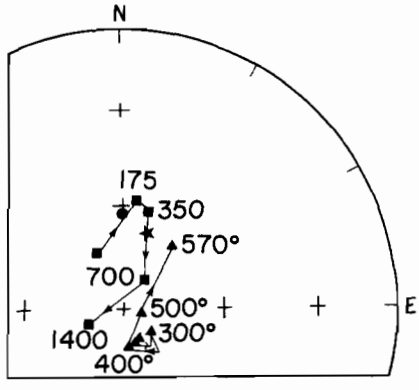
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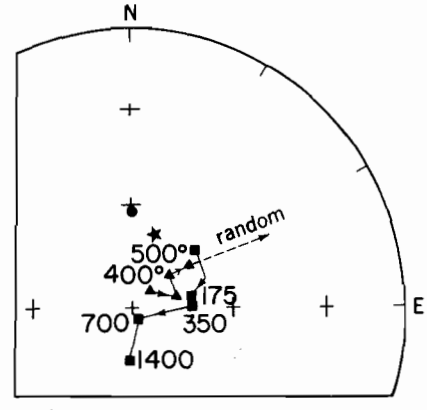
(a) Dike 1



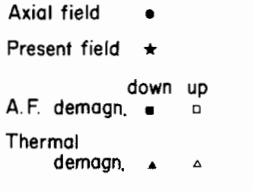
(b) Dike 2

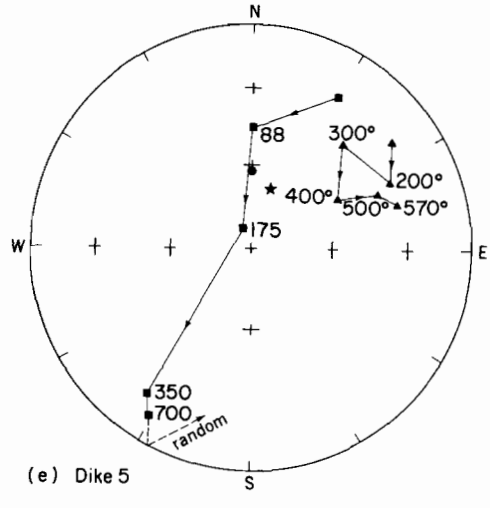


(c) Dike 3



(d) Dike 4





(e) Dike 5

Figure 2. Mean thermal and AF demagnetization paths for each dike, one specimen per sample, plotted on an equal-area projection. Temperatures are in degrees Celsius: fields are in oersted: N is true north. Earth's axial and inclined present magnetic field are also marked.

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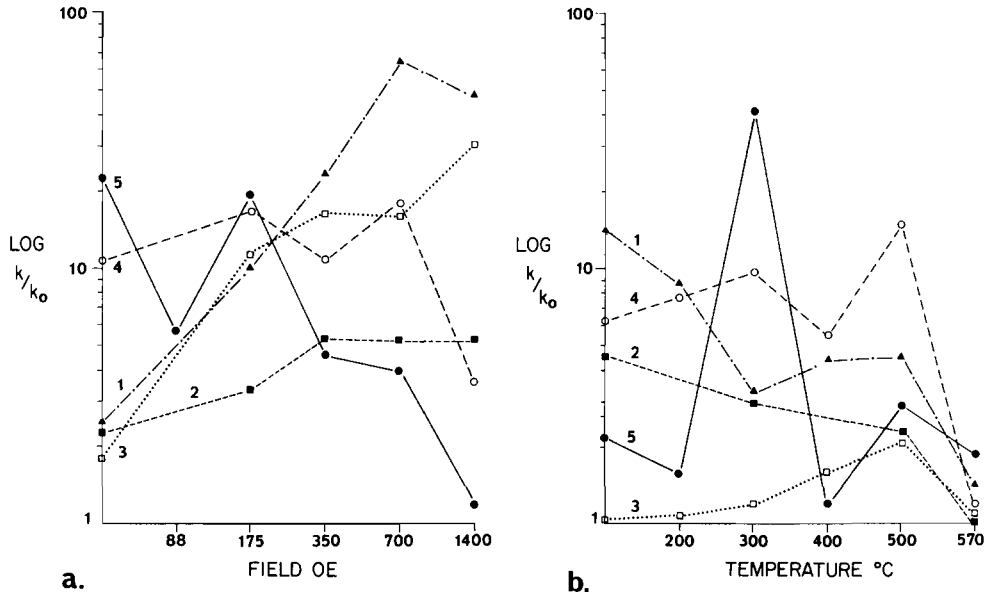


Figure 3. Change of dispersion (measured by log k/k_0) for each dike during progressive thermal and AF demagnetization. The temperature and field ordinates are not to scale, and so there is no implication that the dispersion can be extrapolated to intermediate stages.

creases during thermal demagnetization (Fig. 3b). Yet more than 40 percent of the initial moment of the specimens remains after treatment higher than 500°C.

2. For dike 2, the initial NRM directions also have a shallow inclination, rather similar to those observed at site 1, although they are all of one polarity. Thermal demagnetization produces little change in the mean direction (Fig. 2b) and an increase in the dispersion (Fig. 3b). During AF demagnetization, there is systematic vector rotation toward a vertical inclination (Fig. 2b) and the dispersion decreases (Fig. 3a). The points at which there is minimum dispersion and where systematic vector rotation ends, coincide at 350 oe.

3. In dike 1, a duplicate specimen from one of the samples had an initial shallow inclination, but on AF demagnetization this moved toward a vertical inclination. In addition, one specimen from dike 2 *maintained* a steep inclination during thermal demagnetization, even at high temperatures. None of these specimens was included in computations of the mean, nor have they been included in Figure 2.

4. Several specimens from dike 3 have Q′ ratios near 10 which suggests the effects of lightning (Strangway, 1965). This is further implied by thermal demagnetization because the NRM directions of the individual specimens show little change and no tendency to converge. The apparent lack of change in the *mean* direction (Fig. 2c) is therefore illusory because the dispersion is essentially random at all temperatures except 500°C (Fig. 3b).

5. On AF demagnetization of dike 3 specimens, a systematic vector rotation continues to 1400 oe, at which there is minimum dispersion (Fig. 3a). There is also however, a tight grouping around the present field at 350 oe (Fig. 2c). It demonstrates that large secondary magnetizations acquired in the present field *and* lightning induced isothermal remanent magnetization (IRM) contribute to the over-all remanence, the former persisting to 350 oe. Although the dispersion is minimum at 700 oe, vector rotation continues to 1400 oe, and the dispersion does not change significantly (at the 95 percent level of confidence). The most effective cleaning procedure for this dike is therefore 1400 oe.

6. Specimens from dike 4 are the freshest in hand specimen and give the highest and most consistent apparent K/Ar ages. They also give the most consistent response to *both* types of demagnetization. The AF procedure produces about 30° of vector rotation (Fig. 2d), roughly away from the present field and minimum dispersion occurs at 700 oe. Although the direction changes slightly after the 1400 stage, the

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mean direction at this point cannot be argued as being more reliable because the dispersion is significantly poorer at the 95 percent level (Fig. 3a).

Thermal demagnetization gives 15° of vector rotation which is not related to the present field (Fig. 2d). Minimum dispersion is obtained at 500°C. The mean direction then is 22° away from the mean at 700 oe, but it should be noted that the two initial NRM means *before* demagnetization are 18° apart.

7. Thermal demagnetization of dike 5 specimens produces little direction change (Fig. 2e). There is a remarkably low dispersion at 300°C, yet it is almost random at 400°C. The dispersion at all other temperatures is poor (Fig. 3b), and there is no systematic vector rotation, although there is a more or less anti-clockwise change of 20° from temperature to temperature.

8. The response to AF demagnetization for dike 5 specimens is intriguing (Fig. 2e). The means at 350 and 700 oe are 150° away from the initial direction of NRM, although the vector rotation path passes through the general area of steep inclinations obtained from other dikes. The closest agreement with these other dikes occurs at 175 oe and this is also the stage at which the dispersion is significantly minimum (Fig. 3a). It seems reasonable to suggest that dike 5 shows a partial self reversal phenomenon. All the demagnetization paths for individual specimens follow similar great circles up to 700 oe (passing through the area of steep inclinations from other dikes). Although two of these paths continue on into the upper hemisphere, the dispersion is random at 1400 oe (Fig. 3a). This could be explained as a magnetic noise situation in which any meaningful reversed magnetization is being masked by components of equal or greater magnitude, but having random orientations.

DISCUSSION OF PILOT DEMAGNETIZATION EXPERIMENTS

The main question posed is why two groups of inclinations, steep and shallow, are observed in the initial NRM directions. Three possibilities can be suggested: (1) One of the two groups has been affected by lightning. (2) The two magnetization directions reflect two periods of dike intrusion. (3) An original magnetization has been partially obliterated by another magnetization at a later date.

Lightning can be ruled out because the dike means represent samples collected along as much as a mile of dike length, and it is unlikely that an entire dike would be so completely affected. Furthermore, several samples from dike 3 appear to have been struck by lightning and AF demagnetization removed the IRM acquired.

Over-all, the isotopic evidence provided in an earlier section does not conclusively support the idea of two *distinct* periods of dike injection. Rather it suggested that alteration was responsible for a spread in apparent ages. Note that dike 2 (with a shallow initial inclination) has apparent ages (Table 1) which are intermediate between the apparent ages from dikes 3 and 4 (which have steep inclinations). An apparent age from dike 1 is admittedly much lower (900 m.y.), although this could also be explained in terms of alteration.

For four reasons, it is considered that the dikes possess a primary vertical magnetization, and that this, in two of the dikes, has been dominantly overprinted by a later magnetization:

1. In dike 2, AF demagnetization removes a component with shallow inclination dominant in the initial NRM, and leaves a component with a steep inclination.

2. None of the vertical inclinations shows any tendency to move toward the horizontal on demagnetization.

3. The strongest evidence is provided by a duplicate specimen from dike 1, which on AF demagnetization moves toward the group with steep inclinations, and by a specimen from dike 2 which maintains a steep inclination during thermal demagnetization. This behavior is not due to lightning, because the specimens do not possess either anomalously high intensities or large Q′ ratios.

4. There is some consistency between the initial directions from dikes 1 and 2, which suggests that they may have been magnetized at the same time. This would further imply that the magnetic overprinting was episodic rather than continuous.

OPAQUE MINERALOGY

With the exception of dike 1, all the dikes have a similar Fe-Ti oxide mineralogy in possessing oxidized titanomagnetite, ilmenite, titanomaghemite, and sulfides.

Titanomagnetite occurs as prisms as much as 300 μ long and is typically oxidized to stubby ilmenite lamellae. Some of these are altered to irregular patchworks of anisotropic metailmenite, and these further to hematite and

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rutile. Ilmenite lamellae are developed along both 111 and 110 faces in host titanomagnetite.

The oxidation process seems to have been rather selective. In some grains, the host titanomagnetite remains unaffected; in others, it is altered marginally and along cracks to bluish, quite highly reflecting titanomaghemite, which is accompanied by a submicroscopic granulation. The latter appears to contain hematite (identified purely on the basis of a slight red internal reflection), although Ade-Hall and others, using x-ray analysis (1968), identified a similar granulation in basalts as containing rutile.

There are two types of ilmenite. One occurs in graphic intergrowth with the silicate minerals. The other is skeletal, ranging in size up to 150 × 40 microns, and is oxidized to heterogeneous patchworks of sigmoidal rutile and hematite. All of the Fe-Ti oxides commonly have a partial selvage of translucent sphene around their margins, and there are small amounts of goethite present.

Pyrite and pyrrhotite are abundant, as discrete grains up to 30 μ across, as inclusions in titanomagnetite (and unaffected by the oxidation process) but mainly as a light dusting throughout the whole rock and particularly in the ferromagnesian minerals.

No quantitative analysis of the opaque oxides was made in this study. Rather, the petrography of dikes 3 and 4 (which maintained a steep inclination) was compared with that of dike 5 (which showed partial self-reversal features), and then with that of dikes 1 and 2 (which suggested remagnetization).

In fact, dikes 2, 3, 4, and 5 have a petrography similar to that described above. Dike 5 is distinct in possessing very large, cubic pyrrhotites, as much as 300 μ across.

The distinguishing features of dike 2 specimens are that they have more development of titanomaghemite and granulation, although quantitative estimates of this are difficult. Ade-Hall and others (1968) concluded that such granulation in a basalt flow was a hydrothermal alteration. The presence of titanomaghemite suggests low-temperature alteration, perhaps simply weathering (Katsura and Kushiro, 1961). Desborough (1963) concluded that sphene is formed after solidification of the main body, but that the fluids responsible may be either deuteric or hydrothermal. In the case of the Wind River dikes, the iron liberated by the alteration of titanomagnetite and ilmenite to sphene may well have led to the formation of the sulfides abundant throughout the rock, if the altering solutions contained sulfur. Overall, no conclusions on the *age* of alteration can be put forward, although much of the evidence suggests it was of low-temperature hydrothermal origin.

Dike 1 has a rather different opaque mineralogy. It is much finer grained, and the main minerals are skeletal ilmenites with sphene rims, abundant colloform goethite and a sulfide dusting. Pyrrhotite is inverting along its margin to goethite, and there is a great deal of interstitial alteration to hematite. Some unoxidized titanomagnetites also occur. The interesting fact is that although the opaque mineralogy is different, the silicate mineralogy (in transmitted light) is essentially similar to other dikes.

INTENSITIES, SUSCEPTIBILITIES, AND Q′ RATIOS

Intensities, susceptibilities, and their ratio (Q′) for all the specimens used in the study are listed in Table 3. Because the anomalously high

TABLE 3. INTENSITY, SUSCEPTIBILITY AND Q' RATIO VALUES*

Dike	Intensity x 10^{-4} emu/cc		Susceptibility x 10^{-4} emu/oe/cc		Q' (Intensity/Susceptibility)	
	Range	Median	Range	Median	Range	Median
1	0.1 - 1.2	0.2	0.8 - 1.6	0.9	0.1 - 1.1	0.8
2	0.5 - 116.0	2.2	0.7 - 32.0	1.6	0.1 - 2.8	1.0
3	5.0 - 166.0	73.7	7.8 - 20.4	15.5	0.6 - 13.5	4.5
4	1.1 - 41.2	8.2	0.9 - 27.0	14.6	0.4 - 3.3	0.7
5	0.8 - 19.4	15.3	16.5 - 52.6	41.2	0.1 - 0.4	0.4

* Determined for all Wind River Dike specimens used in this study.

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intensity values from specimens affected by lightning would skew any mean values calculated, the parameters are given in terms of *median* values and their range. Three comments are warranted:

1. The higher median intensity and correspondingly higher Q′ ratios for dike 3 than for other dikes support the idea that several samples have been struck by lightning. This partly explains why thermal demagnetization is ineffective on this dike.

2. Dikes 3, 4, and 5 have similar susceptibility ranges. The intensity/susceptibility ranges from dike 1 are at least an order of magnitude lower than in dikes 3, 4, and 5. Coupled with the microscopic evidence and the very high stability of dike 1 to AF demagnetization, it can be argued that hematite is mainly responsible for the NRM of this dike.

3. Dike 2 has intensity/susceptibility ranges which are midway between those of dike 1 and those of dikes 3, 4, and 5. With the microscopic observation that there is more titanomaghemite and granulation in this dike, it suggests that alteration is the main cause of magnetic variations among the five dikes.

CURIE POINTS

Curie points from five specimens were determined in air in a field of about 2,000 oe. Unfortunately, dike 1 specimens proved too weak, even after mineral separation, to give any kind of meaningful curve. The determinations are shown in Figure 4. They illustrate several points:

1. None of the curves shows a definite Curie point over 600°C, and all are fairly reversible. Thus it was justifiable to omit further thermal demagnetization above 570°C during the pilot experiments.

2. There are two types of curves. One shows a Curie point near 580° to 600°C, and a sharp knee above 550°C. This is typical of a single, *thermally discrete* component of magnetization

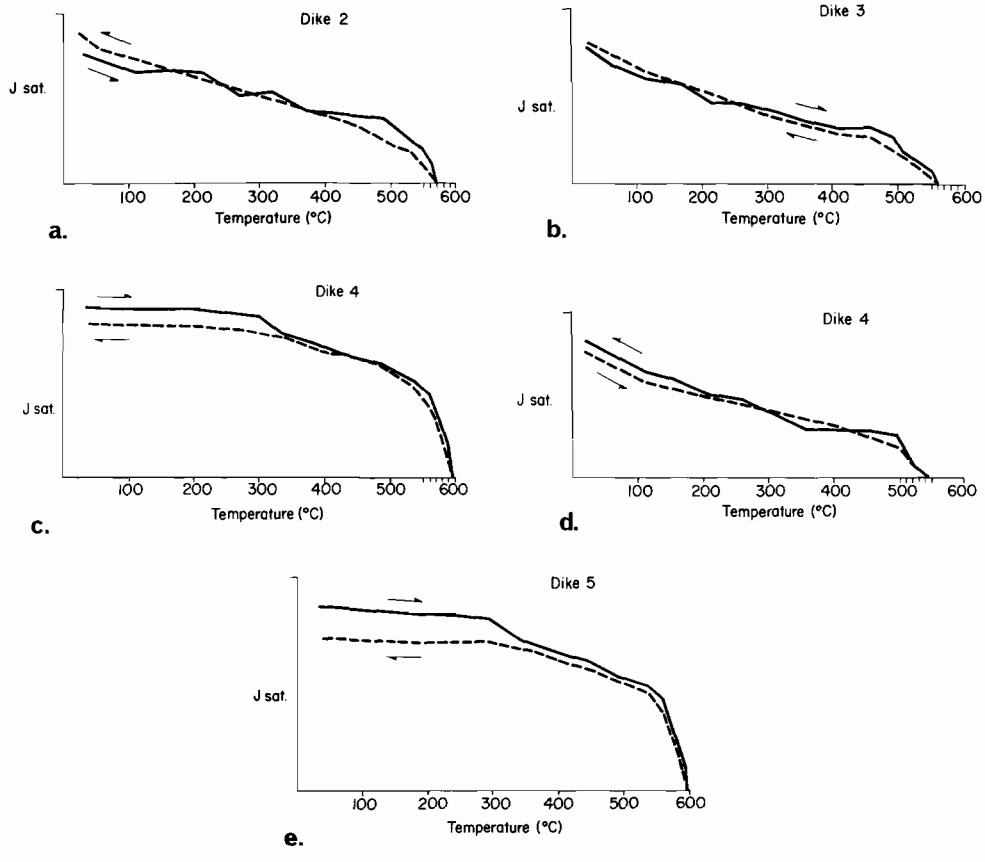


Figure 4. Curie point determinations in a field of about 2,000 oe.

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(Irving and Opdyke, 1965) due to a mineral in the titanomagnetite series (Akimoto, 1954).

The other type of curve shows a more continuous decay, with a lower Curie point in the range 540° to 570°C. The inference here is for a wide range of blocking temperatures (the *thermally distributed* components of Irving and Opdyke, 1965). This is also likely to indicate a mineral in the titanomagnetite series, particularly as a Mössbauer spectrum kindly run by Dr. W. A. Gose on the dike 4 specimen reveals a double Zeeman pattern typical of magnetite.

3. There is a close correlation between the shape of the NRM thermal decay curve and that of the Curie point curve. The *thermally discrete* components present in the curves of Figure 4c and 4e are associated with convex to knee-shaped NRM decay curves. The *thermally distributed* components in the curve in Figure 4d are related to concave NRM decay curves. Note that both types of curve occur within the same dike, supporting the idea that each specimen should be given individual weight.

The conclusion from these few Curie point determinations is that a titanomagnetite is mainly responsible for the saturation remanence, and that there is no apparent contribution from hematite. Ozima and Larson (1967) comment that titanomaghemite is often not shown by a thermomagnetic analysis, so that little can be said about its possible contribution to the saturation remanence.

Two of the specimens show points of inflexion in the interval 300° to 350°C (Figure 4c, 4e). This is particularly so of the dike 5 specimen, and it very probably reflects a contribution from pyrrhotite (Bhimasankaram, 1964).

ARTIFICIAL THERMOREMANENCE

An artificial thermoremanence (TRM) was given to several specimens from each dike, by cooling them from 750°C in a field of 0.3 oe. The angular departure of the TRM from the axis of applied magnetization for all the specimens was less than 3°. This is good evidence that the rocks are magnetically isotropic, and also that any self-reversal feature in dike 5 is not reproducible. Partial thermal demagnetization of most of these specimens gave decay curves which were more convex than the NRM decay curves. Thus further conclusions about whether the NRM is a true TRM are not possible, except to suggest that the rocks are at least capable of acquiring a stable thermoremanence. It is worth noting that for most specimens only a randomly directed moment remained above 530°C, and that the artificial TRM intensity was a factor of 4 to 24 higher than the NRM intensity.

DISCUSSION

The evidence accumulating so far suggests that the steep inclinations represent a primary magnetization, that AF demagnetization is most effective in removing secondary magnetizations, and that during thermal demagnetization the scatter in the NRM directions generally increases. Thus for dikes 2, 3, and 4, blanket AF demagnetization to 350, 1400, and 700 oe, respectively, would give minimum dispersion and no further meaningful vector rotation. Mainly because of the small number of samples and the considerable alteration, no further studies were made on dike 1, although it can be argued that more evidence for steep inclinations might have been found in isolated specimens.

Special mention is required of the dike 5 samples, because it seems likely that pyrrhotite influences their behavior. Anomalous magnetic properties have been encountered in rocks bearing this mineral (Cornwell, 1967), sometimes due to anisotropy (Fuller, 1963). It is possible that for dike 5, the initial NRM directions, remote from the present field (Fig. 2e) and close to the mean direction at 300°C (with its very high precision), reflect the influence of pyrrhotite. In addition, self-reversal features have been observed in pyrrhotite-bearing rocks (Irving, 1964, p. 166-167; Bhimasankaram, 1964), which might explain the response to AF demagnetization (Fig. 2e). However, it must also be pointed out that since the reversed component in dike 5 showed up only in high fields, it implies that it has high coercive force and therefore might be more indicative of hematite. That none was found in the Curie point curve (Fig. 4e) is perhaps due to masking by titanomagnetite.

The magnetic properties of dike 5 are obviously not yet fully understood. Because of its otherwise identical properties with those of the more reliable dikes, the minimum dispersion at 175 oe, and the similarity of the NRM direction at this stage with those from other dikes, it is reasonable at present to say that the data at 175 oe indicate a meaningful direction of magnetization for dike 5, although this field may well have been a fortuitous choice.

The fact that AF demagnetization produces a more consistent response for each dike suggests that stability of remanence in these dikes is

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more related to coercive force than to mineral content, blocking temperature or saturation magnetization. Furthermore, the stable components in each specimen possess a considerable coercivity spectrum. This not only varies within a site (there is little change in the mean NRM direction from site 2 after 350 to 1400 oe treatment) but also between sites (minimum dispersion occurs after 175 oe at site 5, and after 1400 oe at site 3). If titanomagnetite is mainly responsible for the magnetic characteristics of the dikes, and it is assumed that the stable magnetizations reside in single domain particles formed by subdivision of host titanomagnetite into ilmenite lamellae (Strangway and others, 1968), then some speculations can be made about the magnetic noise commonly observed during thermal and high field demagnetization. Irving and others (1961) first proposed that such a residual magnetization was an intrinsic property of some rocks, and Stacey (1963, p. 99-101) showed that it was theoretically possible. In the Wind River dikes, some of the relict titanomagnetite occupies grain areas in excess of 50 microns. Magnetically these will behave as multidomains, and even if their moments became randomly oriented by demagnetization, sufficiently large numbers might become statistically aligned prior to measurement to mask small fractions of a stable primary magnetization remaining after high temperature/field demagnetization. This may explain why a large percentage of the initial intensity often remains above 500°C, although the NRM directions are randomly oriented, and why the dispersion increases significantly in high fields (for example, sites 4 and 5; *see* Fig. 3a).

MEAN DIRECTIONS OF MAGNETIZATION

With the exception of dike 1, duplicate specimens from each sample were AF demagnetized to the stages suggested by the discussion in the previous section. Dike mean directions were computed according to Fisher (1953), giving each specimen individual weight. They are listed in Table 4, with dike 1 data for comparison, as well as the over-all mean combined from dikes 2, 3, 4, and 5, and the corresponding paleomagnetic pole positions. Application of Hartley's maximum F ratio test on the four dispersion parameters (k) at the 95 percent level of confidence (Watson and Irving, 1957) suggests that the dike directions are drawn from populations with dissimilar dispersions, which is in agreement with the variable response of each dike to demagnetization. The greatest dispersion (lowest k value) occurred in dike 2, and this must be due to remaining amounts of the secondary magnetization which was very obvious in the initial NRM. Figure 5 shows the four dike means and their over-all mean, with the associated circles of 95 percent confidence. It is immediately obvious that the dike 2 mean is central to the other dike means, and thus apparently not magnetically different. The dispersion of the over-all mean $\Delta = 81\ k^{-1/2}$ is 10.1°, which suggests (*after* Creer, 1962) that no large errors due to secular variation are present.

PALEOMAGNETIC POLES

The corresponding paleomagnetic poles (Table 4) are drawn in Figure 6 with their respective ovals of 95 percent level confidence. They

TABLE 4. MEAN DIRECTIONS OF MAGNETIZATION AND PALEOMAGNETIC POLES FOR WIND RIVER DIKES

Dk	Fld	Dec	Inc	N	R	k	α_{95}	Long	Lat	δp	δm
1	0	290	1	3	2.844	13	36	148E	15N	18	36
1	700	287	0	3	2.994	357	6	149E	12N	3	6
2	0	250	53	10	7.000	3	34	145E	35S	32	46
2	350	269	86	10	9.497	18	12	121W	42N	23	23
3	0	331	79	10	7.947	4	26	128W	59N	47	49
3	1400	264	73	10	9.895	86	5	146W	32N	8	9
4	0	39	73	12	11.347	17	11	66W	61N	17	19
4	700	112	84	12	11.760	46	6	95W	37N	12	13
5	0	29	31	14	13.328	19	9	17E	53N	6	10
5	175	342	84	14	13.573	30	7	115W	54N	14	14
Mean of 2, 3, 4 and 5		277	85	4	3.954	65	11	121W	43N	23	23

Dk is dike number; Fld is alternating field in oersted; Dec is declination of the north-seeking magnetic vector, east of true north; Inc is its inclination, positive downward; R is the length of N unit vectors, one per specimen; k = (N-1) / (N-R); α_{95} is the semi-angle of the cone of 95% level confidence; Long and Lat are the longitude and latitude respectively of the paleomagnetic pole; δp and δm are the semi-minor/major axes of the oval of 95% level confidence.

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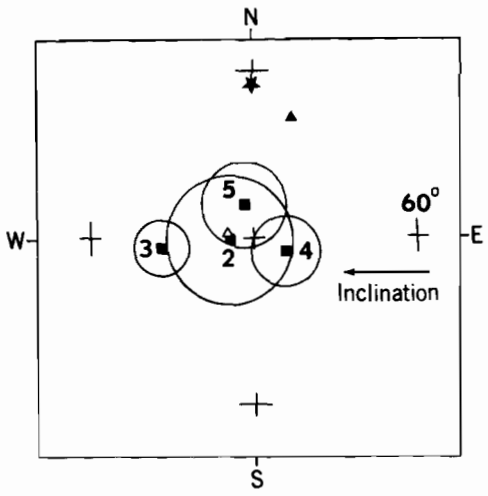


Figure 5. Mean directions of magnetization for dikes 2, 3, 4, and 5 plotted on an equal area projection (all directions downward seeking), with their associated circles of 95 percent level confidence. Over-all mean direction is given by the open triangle. Asterisk is the axial field, and solid triangle is the inclined present field.

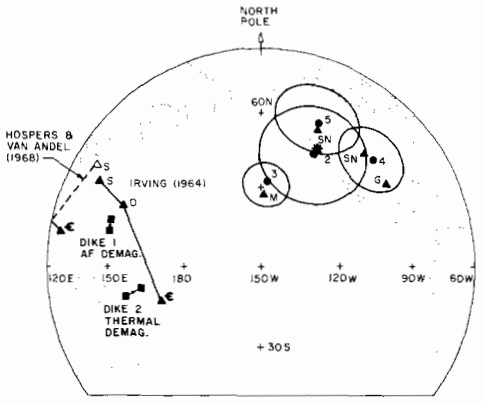


Figure 6. Paleomagnetic poles for dikes 2, 3, 4, and 5 plotted on an equatorial area projection with their associated ovals of 95 percent level confidence. Asterisk is the over-all mean pole. Triangles are other Precambrian poles (*see* text) for the Sudbury Nickel Irruptive (SN), Marathon dikes (M), and Gunflint Iron Formation (G). Demagnetization paths for dike 1 (AF) and dike 2 (thermal) are also shown with mean Lower Paleozoic pole paths for North America by Irving (1964) and Hospers and Van Andel (1968). € is Cambrian, O is Ordovician, S is Silurian: dotted line indicates poles on the opposite hemisphere.

all lie in the northeastern Pacific region and are within the locus of several other poles from rocks of similar age from North America.

Two structurally corrected poles are available for the Sudbury Nickel Irruptive (Hood, 1961; Sopher, 1963), which is dated at 1704 m.y. (Fairbairn and others, 1968); they are labelled SN in Figure 6. Another pole (M) is from the 1810-m.y.-old Marathon dikes (Fahrig and others, 1965). A fourth pole (G) is from the Gunflint Iron Formation (Symons, 1966) which is associated with sediments of the Animikie Series. The absolute age of the Animikie Series is uncertain. Goldich (1968) notes that isochrons of 1540, 1650, and 1850 m.y. have been obtained for these rocks, and that the youngest age may be related to some form of alteration. Faure and Kovach (1969) consider that an Rb/Sr isochron age of 1635 m.y. from the Gunflint Iron Formation may well represent the time of diagenesis of the series in Ontario. Thus it seems reasonable to conclude that the *pole* from the Gunflint Formation represents some part of the interval from 1600 to 1850 m.y.

There are exceptions to this coherent group of poles. Larochelle (1969) points out certain limitations in the data from the Sudbury body. Symons (1967) reports two poles from the Negaunee Iron Formation (assigned to the Animikie Series) which are not only remote from the poles discussed above, but differ from each other by 70° of arc. The problem here is not simply *when* were these iron ores magnetized but *how* were they magnetized. Obviously more poles from well-dated rock units would materially help in charting the paleofield axis during this interval of Precambrian time. Nevertheless, Figure 6 shows that there is good agreement between poles from four isotopically dated rock units, of different lithology and, more important, from different parts of the North American Shield. They provide evidence that the Precambrian field axis in the interval 1600 to 2060 m.y. was in the vicinity of 95° to 150° W. and 30° to 50° N.

AGE OF REMAGNETIZATION

For comparison, the AF demagnetization path for dike 1, and the thermal demagnetization path for dike 2 are also shown in Figure 6. Both sets of poles lie in the western Pacific near the equator, but they differ by about 30°, thus raising uncertainty as to whether in fact they represent the same age of remagnetization. Evidence for the same magnetic age is given by these facts: (1) both dikes have the same trend and gross petrology (Nold, 1964), and (2) a

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specimen from dike 1 during AF demagnetization also reveals the steep inclination which is common to all the other dikes. Evidence that the dikes were remagnetized at different times is supported by the following points: (A) the discrepancy between the pole positions, (B) the mixed polarities in dike 1, contrasted with the single polarity in dike 2, (C) the different opaque mineralogy of dike 1, and (D) the very low apparent isotopic age (900 m.y.) from dike 1.

Further studies on *all* the dikes from the southern Wind River Range are in progress in an attempt to resolve this problem. At present the simplest explanation is put forward, namely that the two dikes were remagnetized at the same time. Several pieces of evidence are relevant to the date of this overprinting:

1. A whole-rock K/Ar age of 900 m.y. is reported for dike 1 by Condie and others (1969).

2. The two sets of pole positions for these dikes show more affinity for Lower Paleozoic poles from North America than for Precambrian poles (*see* summary of the latter by Spall, 1971). In Figure 6, two mean pole paths for North American Lower Paleozoic rocks have been redrawn from Irving (1964) and Hospers and Van Andel (1968). The discrepancy between these two paths is due to two reasons; first, the paucity of data, and second, the dispersion among the few poles available (for an example of this, *see* Helsley and Spall, 1966). However, new studies by the author on Lower Paleozoic rocks from Virginia and dated intrusives from Colorado (E. E. Larson, 1971, personal commun.) confirm the broad outline of these two curves, so that it is justifiable to use them for approximate dating purposes.

3. All five dikes are overlain by upward of 2,500 ft of Paleozoic section, of which 1,200 ft are assigned to the Cambrian, (Branson and Branson, 1941; Keefer and Van Lieu, 1970).

All of this evidence implies that remagnetization of both dikes occurred at either 900 m.y. (and the whole-rock age is a true age), or during the Lower Paleozoic (probably Cambrian), or at some time between 2,000 m.y. and the Lower Paleozoic, when the rocks were close to an erosion surface. With the evidence available, the simplest explanation is that Cambrian remagnetization is responsible for the magnetic behavior of dikes 1 and 2. Whether this is related to the events which reset the isotopic clocks cannot be determined, since no Phanerozoic apparent ages have been reported for Precambrian rocks in Wyoming. It is curious that dike 1 should be so completely remagnetized when it is no nearer to the Cambrian erosion surface than any other dike (Fig. 1).

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